DERIVED PUSHFORWARD ON QUASI-COHERENT COMPLEXES COMMUTES WITH DIRECT SUMS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For a quasi-compact quasi-separated morphism the pushforward on quasi-coherent complexes has finite cohomological dimension N, so R^0f_* agrees with R^0f_* of the truncation tau_{>= -N}. used at proof lines 817-822. Lets every family be truncated to a uniformly bounded below one.
- The cohomology of the pushforward of a complex is computed from the pushforwards of its cohomology sheaves by the two spectral sequences of a complex functor. invoked at proof lines 824-830. Reduces the general direct sum statement to the case where the objects are sheaves in a single degree.
- For quasi-coherent sheaves the cohomology of a direct sum is the direct sum of the cohomologies on a quasi-compact quasi-separated scheme. invoked at proof lines 831-835. Handles the single-degree case and finishes the proof.

Source reference. Original lemma 08D5: sources/stacks-perfect/pinned-original-source.tex, source lines 661–661. Cited at proof lines 817-820 to choose N and justify the truncation; the uniform truncation of the family is inside the counted span.

Source reference. Original lemma 015J: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 824-830 to reduce to sheaves; the use of both spectral sequences for the family is inside the counted span.

Source reference. Original lemma 07TB: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 831-835 to handle the single-degree case; the assembly of the two reductions is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a quasi-separated and quasi-compact morphism of schemes. Then $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_S)$ commutes with direct sums.*

Proof. Let E_i be a family of objects of $D_{QCoh}(\mathcal{O}_X)$ and set $E = \bigoplus E_i$. We want to show that the map

$$\bigoplus Rf_*E_i \longrightarrow Rf_*E$$

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is an isomorphism. We will show it induces an isomorphism on cohomology sheaves in degree 0 which will imply the lemma. To prove this we may work locally on S, hence we may and do assume that S is quasi-compact. Choose an integer N as in Lemma 08D5. Then $R^0 f_* E = R^0 f_* \tau_{\geq -N} E$ and $R^0 f_* E_i = R^0 f_* \tau_{\geq -N} E_i$ by the lemma cited. Observe that $\tau_{\geq -N} E = \bigoplus \tau_{\geq -N} E_i$. Thus we may assume all of the E_i have vanishing cohomology sheaves in degrees $< -N$. Next we use the spectral sequences

$$R^p f_* H^q(E) \Rightarrow R^{p+q} f_* E \quad \text{and} \quad R^p f_* H^q(E_i) \Rightarrow R^{p+q} f_* E_i$$

(Derived Categories, Lemma 015J) to reduce to the case of a direct sum of quasi-coherent sheaves. This case is handled by Cohomology of Schemes, Lemma 07TB. $\square$